

THANH HOA, Viet Nam

WMO: 488400

Lat: 19.750N

Lon: 105.783E

Elev: 5

StdP: 101.26

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.9	12.0	4.9	5.4	14.0	6.6	6.0	14.2	7.5	20.6	6.5	20.2	2.0	0	0.283

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	5.9	35.7	27.5	34.5	27.6	33.4	27.5	29.0	32.5	28.6	32.1	28.2	31.6	2.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.1	24.3	31.2	27.7	23.6	30.7	27.3	23.1	30.4	94.9	32.6	92.9	32.3	91.1	31.8	33.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.4	5.5	4.7	DB	9.6	37.9	1.5	1.1	8.5	38.8	7.6	39.4	6.7	40.1	5.6	40.9	(4)
(5)			WB	7.5	30.0	1.4	0.9	6.5	30.6	5.6	31.2	4.8	31.7	3.8	32.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	24.4	17.6	18.6	20.6	24.5	27.8	30.0	29.6	28.7	27.6	25.8	22.8	19.1	(6)
(7)		DBStd	4.99	3.21	3.54	2.93	2.67	2.29	1.94	1.70	1.53	1.64	1.90	2.61	2.90	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	130	52	38	12	1	0	0	0	0	0	0	2	26	(9)
(10)		CDD10.0	5263	237	239	329	434	553	599	607	581	528	489	383	282	(10)
(11)		CDD18.3	2351	31	44	83	184	295	349	348	323	278	231	134	49	(11)
(12)		CDH23.3	22791	27	74	194	1168	3161	4551	4405	3785	2866	1809	649	103	(12)
(13)		CDH26.7	8558	1	7	15	226	1173	2237	2011	1473	922	404	85	4	(13)
(14)	Wind	WSAvg	1.7	1.8	1.7	1.6	1.8	1.8	1.9	1.8	1.5	1.6	1.8	1.9	1.7	(14)
(15)	Precipitation	PrecAvg	1636	26	16	45	61	150	163	203	282	367	212	81	30	(15)
(16)		PrecMax	2390	123	47	142	203	317	410	416	664	707	644	480	91	(16)
(17)		PrecMin	1053	3	2	5	16	34	53	73	64	80	26	5	4	(17)
(18)		PrecStd	309	27	12	33	43	81	80	86	134	154	160	93	24	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.3	27.3	28.5	33.3	36.8	37.5	36.9	35.5	33.9	31.6	29.8	27.2	(19)
(20)			MCWB	22.0	23.2	24.7	27.2	27.5	27.3	27.7	27.9	27.4	25.2	25.1	23.4	(20)
(21)		2%	DB	24.1	25.2	26.5	30.5	34.4	35.9	35.0	34.0	32.5	30.6	28.7	25.7	(21)
(22)			MCWB	21.4	22.5	24.0	26.6	27.5	27.4	27.8	28.0	27.2	25.1	24.4	22.5	(22)
(23)		5%	DB	22.9	24.1	25.3	28.9	32.6	34.6	33.8	32.7	31.6	29.8	27.7	24.4	(23)
(24)			MCWB	20.8	22.2	23.4	26.0	27.5	27.3	27.7	27.7	26.8	24.9	23.7	21.9	(24)
(25)		10%	DB	21.9	23.2	24.3	27.8	31.2	33.4	32.6	31.7	30.7	28.9	26.6	23.2	(25)
(26)			MCWB	20.1	21.6	22.8	25.4	27.2	27.3	27.6	27.5	26.4	24.7	23.3	20.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.9	23.8	25.3	28.2	29.2	29.3	29.7	29.3	28.7	27.3	26.2	24.5	(27)
(28)			MADB	24.4	25.8	27.4	31.4	32.7	33.1	33.4	32.6	31.7	29.6	28.2	26.2	(28)
(29)		2%	WB	22.0	23.0	24.4	27.0	28.5	28.7	28.8	28.7	28.1	26.5	25.5	23.4	(29)
(30)			MADB	23.3	24.5	26.0	29.8	32.0	32.5	32.4	32.0	31.0	28.9	27.5	24.9	(30)
(31)		5%	WB	21.2	22.5	23.6	26.3	28.0	28.3	28.4	28.2	27.6	26.0	24.9	22.5	(31)
(32)			MADB	22.5	23.8	25.0	28.4	31.5	32.0	31.9	31.4	30.3	28.2	26.8	23.9	(32)
(33)		10%	WB	20.4	21.9	23.0	25.7	27.5	27.9	28.0	27.7	27.1	25.5	24.2	21.6	(33)
(34)			MADB	21.6	23.1	24.1	27.4	30.6	31.5	31.4	30.8	29.8	27.7	26.0	23.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.2	3.9	4.0	4.5	5.6	5.9	5.5	5.4	5.2	5.1	5.3	4.9	(35)
(36)			MCDBR	5.6	5.1	5.1	6.1	7.8	7.6	7.2	7.2	6.6	6.3	6.2	6.2	(36)
(37)			MCWBR	2.8	2.7	2.9	3.3	3.4	2.9	2.9	3.1	2.7	2.2	2.7	3.1	(37)
(38)		5% WB	MCDBR	4.8	4.5	4.8	5.8	6.7	6.4	6.3	6.4	6.0	5.2	5.2	5.4	(38)
(39)			MCWBR	2.9	2.6	2.9	3.4	3.5	3.1	3.1	3.1	2.9	2.4	2.8	3.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.597	0.632	0.781	0.736	0.552	0.495	0.468	0.518	0.586	0.692	0.605	0.635	(40)	
(41)		taud	1.762	1.733	1.497	1.613	2.070	2.249	2.318	2.157	1.934	1.663	1.800	1.708	(41)	
(42)		Ebn at Noon	679	687	608	645	768	806	829	793	735	633	668	629	(42)	
(43)		Edh at Noon	211	228	299	268	168	140	130	154	190	241	201	217	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.35	2.85	3.12	4.24	5.31	5.40	5.14	4.73	4.32	3.71	3.25	2.46	(44)	
(45)		RadStd	0.51	0.48	0.38	0.41	0.48	0.41	0.56	0.36	0.39	0.36	0.41	0.39	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	+0.50	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		+0.30	N/A	N/A	+0.60	N/A	N/A	N/A	N/A	N/A	+110

Nomenclature: See separate page